

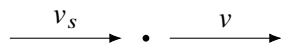
Notebook 2012 — Pagina's 65, 66 en 69–72

Pagina's 67 en 68 zijn op verzoek overgeslagen. Deze batch bevat vooral formulebladen, korte definities en een Doppler-afleiding. Minder scherpe symbolen zijn zo letterlijk mogelijk overgenomen en als onzeker gemarkeerd waar nodig.

Pagina 65

Linkerhelft

Frequentie van een object wordt
20% hoger voor een observator.



$$N_1 = \left(\frac{V}{V - v_s} \right) N \quad = \text{op je af}$$

$$N_1 = \left(\frac{V}{V + v_s} \right) N \quad = \text{van je af}$$

$$N_1 = \left(\frac{V}{V - v_s} \right) n$$

$$\frac{V - v_s}{V} = \frac{n}{n_1} = \frac{100\%}{120\%} = \frac{5}{6}$$

$$1 - \left(\frac{v_s}{V} \right) = \frac{5}{6}$$

$$\frac{v_s}{V} = 1 - \frac{5}{6} = \frac{1}{6}$$

$$\frac{V + v_s}{V} = \frac{n}{n_2}$$

$$1 + \left(\frac{v_s}{V} \right) = \frac{n}{n_2}$$

$$1 + \frac{1}{6} = \frac{n}{n_2}$$

$$\frac{7}{6} = \frac{n}{n_2}$$

$$n_2 = \frac{6}{7}n$$

$$\frac{n - n_2}{n} \times 100 = \left(1 - \frac{6}{7} \right) \times 100 \approx 14,3\%$$

Rechterhelft

700 Hz



$$N_1 = \left(\frac{V + v_0}{V - v_s} \right) N$$

Hoekmoment = mVR

Linkerhelft

$$R_{\infty} = \frac{\alpha^2 m_e c}{4\pi\hbar} = \frac{\alpha^2}{2\lambda_e} = \frac{\alpha}{4\pi a_0}$$

$$hcR_{\infty} = m_e c^2 \frac{\alpha^2}{2} = \frac{hc \alpha^2}{2\lambda_e} = \frac{hf_c \alpha^2}{2}$$

$$\frac{\hbar\omega_c}{2} \alpha^2 = \frac{\hbar^2}{2m_e a_0^2} = \frac{e^2}{(4\pi\epsilon_0) 2a_0}$$

De eerste noemer is geschreven als $4\pi\hbar$, wat equivalent is aan $2h$.

Rechterhelft

$$\lambda_e = \text{Compton golflengte} = \frac{h}{m_e c}$$

$$f_c = \text{Compton frequentie} = \frac{m_e c^2}{h}$$

$$\omega_c = \text{Compton moment} = 2\pi f_c$$

Pagina 69

Linkerhelft

Bron:

minim JavaScript

Daniel Tammet → hoofd rekenen

Rechterhelft

Snelheid van fonon

$$V = f \times \lambda$$

$$f = \frac{1}{2\pi} \omega$$

$$f = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$$

$$V = \frac{1}{2\pi} \sqrt{\frac{K}{m}} \lambda$$

$$V = \frac{1}{2\pi} \sqrt{\frac{F_{\max}}{2R_p m_p}} (2R_c)$$

$$V = \left(\frac{R_c}{\pi} \right) \sqrt{\frac{F_{\max}}{2R_p m_p}}$$

Pagina 70

Linkerhelft

$$E = hf$$

$$C_e = f\lambda$$

$$c = \frac{\varepsilon_0 A}{D}$$

$$E = \frac{Q^2}{2C}$$

fotoElektrisch effect

$$(1) \quad C_e = f\lambda \quad \Rightarrow \quad \lambda = \frac{C_e}{f}$$

$$(2) \quad c = \frac{\varepsilon_0 A}{D} \quad \Rightarrow \quad c = \frac{\varepsilon_0 \lambda^2}{0.5 \lambda}$$

$$(3) \quad c = \frac{\varepsilon_0 \lambda^2}{0.5 \lambda} \quad \Rightarrow \quad c = 2\varepsilon_0 \lambda$$

$$(4) \quad c = 2\varepsilon_0 \lambda \quad \Rightarrow \quad c = \frac{2\varepsilon_0 C_e}{f}$$

$$(5) \quad E = \frac{Q^2}{2C} \quad \Rightarrow \quad E = \frac{Q^2 f}{4\varepsilon_0 C_e}$$

ε_0 = max amplitude van het diëlektrum

hier van het vacuüm (tijdruimte)

Rechterhelft

$$f_c = \frac{m_e c^2}{h} \quad \omega_c = 2\pi f_c$$

$$V = \omega_e R_c \quad \omega_e = \sqrt{\frac{K_e}{m_e}} \quad K_e = \frac{F_{\max}}{R_x}$$

Vrije vortex

$$C_e = C_i R_e N$$

$$C_e = \sqrt{\frac{K_e}{m_e Z}} R_c N$$

$$C_e = \sqrt{\frac{F_{\max}}{R_x m_e Z}} R_c N$$

$$C_e^2 = \frac{F_{\max}}{R_x m_e Z} R_c^2 N^2$$

$$R_x = N^2 \left[\frac{F_{\max} R_c^2}{m_e C_e^2 Z} \right]$$

Basis formules

elektronen Radius

$$a_0 = \frac{F_{\max} R_c^2}{m_e C_e^2}$$

$$a_0 = h \cdot \frac{c^2}{8\pi F_{\max} C_e R_c}$$

$$a_0 = \frac{c^2 R_c}{2C_e^2}$$

$$\frac{1}{R_c} = \frac{c^2}{a_0 2C_e^2}$$

$$a_0 = \frac{h}{4\pi m_e C_e}$$

$$h = 4\pi m_e C_e a_0$$

$$h = \frac{4\pi F_{\max} R_c^2}{C_e}$$

$$\alpha = \frac{2C_e}{c}$$

$$m_e = \frac{2F_{\max} R_c}{c^2}$$

$$R_x = n^2 \left(\frac{F_{\max} R_c^2}{m C_e^2 Z} \right)$$

$$R_{\infty} = \frac{m_e c \alpha^2}{2h}$$

$$\frac{C_e e^2}{8\pi \varepsilon_0 R_c^2 c F_{\max}} = \alpha$$

$$R_e = 2R_c$$

$$R_e = \frac{e^2}{4\pi \varepsilon_0 m_e c^2}$$

$$a_0 = \frac{\alpha}{4\pi R_{\infty}}$$

$$\alpha_g = \frac{G m_e^2}{\hbar c} = \left(\frac{m_e}{m_{\text{planck}}} \right)^2 = (t_p f_c)^2$$

$$e = \sqrt{2\alpha h \sqrt{\mu_0 c_0}}$$

$$\text{Hoekmoment} = m v_e$$

Pagina 72

Linkerhelfft

Formules

$$E = mc^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)$$

$$E = hf$$

$$E = \frac{1}{2}Kx^2$$

$$E = \frac{2q^2}{4\pi\epsilon_0} \cdot \frac{1}{x}$$

$$\omega = \sqrt{\frac{K}{m}}$$

$$\omega = 2\pi f$$

$$f = \frac{1}{2\pi}\omega$$

$$V = f\lambda$$

$$K = \frac{F_{\max}}{x}$$

$$K = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

$$\lambda = \frac{h}{p}$$

$$C_e = \frac{c\alpha}{2}$$

$$C_e = \omega R_c$$

$$a_0 = \frac{c^2 R_c}{2C_e^2}$$

$$a_0 = \frac{F_{\max} R_c^2}{m_e C_e^2}$$

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}$$

$$a_0 = \frac{\hbar}{m_e c \alpha}$$

$$E = \frac{1}{2}mv^2$$

$$W = \vec{F} \cdot \vec{x}$$

$$F = ma$$

$$a = \frac{v}{t}$$

$$v = \frac{s}{t}$$

Rechterhelfft

$$\omega = mg$$

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$t' = \frac{t - \frac{v}{c^2}x}{\sqrt{1 - v^2/c^2}}$$

$$\lambda_e = \frac{h}{m_e c}$$

$$f_c = \frac{m_e c^2}{h}$$

$$\omega_c = 2\pi f_c$$

$$E_e = \frac{4\pi F_{\max} R_c^2}{C_e} \cdot f_e$$

$$a = -\omega^2 x$$

$$a = -4\pi^2 f^2 x$$